

Ankit Sharma



1. Personal Details

Surname	Sharma
First Name	Ankit
Email	a_sharma@wr.iitr.ac.in
ORCID	0009-0007-4294-0753
Date of CV	06 June 2026
Affiliation	Indian Institute of Technology (IIT) Roorkee, Roorkee, Uttarakhand, India

2. Research Profile

My research combines satellite Earth observation, field measurements, machine learning, and deep learning to investigate water and agricultural systems from local to global scales. I develop approaches for inland water mapping and classification, water storage assessment, quantifying trade-offs in inland water resources, and satellite-based agricultural monitoring.

3. Selected Research Contributions

- First global framework for mapping and classifying lentic and lotic inland waters down to 0.1 ha.
- Global inventory of ~95 million inland waterbodies, highlighting the overlooked contribution of small inland waterbodies to freshwater storage.
- Storage and Reachability Index (StoRI), demonstrating the importance of water reachability alongside storage availability.
- First global classification of inland waterbody types and their environmental controls.
- Four-decade assessment of small-waterbody storage dynamics and their potential for basin-scale flood mitigation via TOI (Tankshed Overflow Index).
- Co-developed IndiaRice500, a long-term gridded rice yield dataset for India derived from self-supervised digital twins, integrating satellite Earth observation with climate and management datasets.
- Co-developed CropLizer, a farm-specific digital advisory system that supports fertilizer and water decisions for smallholder rice farmers, demonstrating real-world translation of research.

4. Technical Skills

Programming	Python and R (primary), JavaScript for Google Earth Engine
Geospatial	ArcGIS, QGIS, ERDAS IMAGINE
Computing	PARAM Ganga HPC cluster
Hydrological Modelling	HEC-RAS (1D and 2D, dam-breach and inundation), HEC-HMS, SWAT, SNAP

5. Earth Observation Experience

- Long-term Landsat time-series analysis (1988–present)
- Sentinel-based surface-water mapping
- ICESat-2 photon lidar applications
- UAV data collection (Trinity F90+)

- Geospatial analysis via Google Earth Engine
- Integration of satellite observations with field measurements

6. Education

PhD Thesis Submitted (May 2026)	Doctor of Philosophy (PhD) Major: Water Resources Engineering Department: Water Resources Development and Management Institute: Indian Institute of Technology (IIT), Roorkee, Uttarakhand, India Thesis: Emerging Role of Small Inland Waterbodies in Storage Dynamics Advisor: Dr. I. Idhayachandhiran
2020	Master of Technology (MTech) Major: Environmental Engineering Department: Center for Energy and Environmental Engineering Institute: National Institute of Technology (NIT), Hamirpur, Himachal Pradesh, India Thesis: Hydrodynamic Modeling of Glacial Lake Outburst Flood (GLOF) in Manalsu River and its impact on Manali Town Advisor: Dr. Chander Prakash
2017	Bachelor of Engineering (BE) Major: Civil Engineering Department: Civil Engineering Institute: Chandigarh University, Mohali, Chandigarh, India

Professional Training

- Remote sensing and satellite altimetry for inland waterbodies (IIRS)
- GRACE data applications (IIT Kanpur)
- Advanced tools and techniques for hydrological investigation (NIH)
- ERDAS IMAGINE (Hexagon Geospatial)
- TUFLOW hydraulic modelling
- R programming for hydro-climatic modelling (IIT Roorkee)

7. Current Employment

Date	Aug 2020 – Present
Title	Doctoral Researcher (Full-time PhD Student)
Department and Institution	Department of Water Resources Development and Management, Indian Institute of Technology (IIT) Roorkee, Uttarakhand, India
Degree Program	Ph.D. in Water Resources Engineering
Academic Career Stage	Stage I (Early-stage researcher- Doctoral Level)
Description	Conducting doctoral research using satellite Earth observation, machine learning, and field observations to quantify inland-water storage dynamics across local to global scales. Research activities include large-scale geospatial analysis, ML/DL model development and validation, scientific publishing and conference presentations, and supervision of student research projects.

8. Publications

A. Peer-reviewed Journal Publications (Published)

Sharma, A., & Ilampooranan, I. (2026). Small and medium-sized inland waterbodies: Water volume predictions and flood implications. *Water Resources Research*, 62(1), e2024WR038283. (Q1; impact factor = 5; h index = 290).

Sharma, A., Narayanan, M., & Ilampooranan, I. (2026). An automated data efficient morphometric approach to define global lentic and lotic inland waters. *Water Resources Research*, 62(1), e2025WR040137. (Q1; impact factor = 5; h index = 290).

Sharma, A., & Ilampooranan, I. (2023). Mapping and monitoring inland water bodies: A case study of Madurai, Tamil Nadu. In *IGARSS 2023–2023 IEEE International Geoscience and Remote Sensing Symposium* (pp. 2894-2897). IEEE. (h index = 94).

B. Book Chapter

Narayanan, M., **Sharma, A.,** & Ilampooranan, I. (2025). Precision agriculture and water management in India: Artificial intelligence for climate action. In *Integrated land and water resource management for sustainable agriculture* (Vol. 1, pp. 17–40). Springer Nature.

C. Peer-reviewed Journal Publications (In preparation)

Sharma, A., Ilampooranan, I., Narayanan, M., Messenger, M., Rajagopalan, B., & Lauerwald, R. (2026). Small lakes drive reachability as large lakes dominate global storage. [Manuscript in preparation].

Sharma, A., Ilampooranan, I., & Narayanan, M. (2026). Types and controls of global inland waters. [Manuscript in preparation].

Eruventi, N.[†], **Sharma, A.[†]**, Narayanan, M., & Ilampooranan, I. (2026). An unsupervised multidimensional clustering framework for mapping inland waters. [Manuscript in preparation].

[†] Equal contribution.

D. Peer-reviewed Journal Publications (Co-authored, under review)

Narayanan, M., Ilampooranan, I., & **Sharma, A.** (2026). Farm-specific digital advisories mitigate fertilizer overuse and double profitability for smallholding rice farmers [Preprint]. SSRN. <https://ssrn.com/abstract=6027854>. [Under Review].

Narayanan, M., Ilampooranan, I., McKeown, P. C., **Sharma, A.,** Ndlovu, N., & Spillane, C. (2026). Water-efficient Indian rice cultivation boosts exports despite high carbon footprints [Preprint]. EarthArXiv. <https://doi.org/10.31223/X5DN3T>. [Under Review].

Narayanan, M., **Sharma, A.,** & Ilampooranan, I. (2025). Spatio-temporally explicit decoupling of rice yield and methane emission relationships in India [Preprint]. SSRN. <https://ssrn.com/abstract=5781084>. [Under Review].

Narayanan, M., **Sharma, A.,** & Ilampooranan, I. (2025). IndiaRice500: Annual gridded rice yield dataset for India from self-supervised digital twins during 2004 to 2021. [Under Review].

E. Conference Presentations and Abstracts

Sharma, A., Narayanan, M., & Ilampooranan, I. (2026). An automated morphometric approach for global lentic and lotic classification of inland waters. *EGU General Assembly 2026*.

Sharma, A., & Ilampooranan, I. (2025). Declining water storage in small and medium inland waterbodies of Chennai: Implications for mitigating flood risks. *IAHS Scientific Assembly 2025*.

Sharma, A., & Ilampooranan, I. (2024). Water volume prediction and flood implications of small and medium inland waterbodies (Abstract H06-85). *AGU Fall Meeting 2024*.

Sharma, A., Narayanan, M., & Ilampooranan, I. (2024). Advancing inland waterbody mapping with self-supervised machine learning (Abstract 16879). *EGU General Assembly 2024*.

Narayanan, M., **Sharma, A.,** & Ilampooranan, I. (2024). Rice production links to methane emissions: Insights from mapping rice areas and yields using self-supervision (Abstract GC34E-04). *AGU Fall Meeting 2024*.

Narayanan, M., **Sharma, A.,** & Ilampooranan, I. (2024). Self-supervised learning for mapping rice areas reduces the need for field surveys (Abstract 16460). *EGU General Assembly 2024*.

Sharma, A., Narayanan, M., & Ilampooranan, I. (2023). Beyond manual labeling: A self-supervised framework for mapping inland water bodies (Abstract H52G 07). *AGU Fall Meeting 2023*.

Sharma, A., Narayanan, M., & Ilampooranan, I. (2022). Augmenting data in scarce regions helps accurately upscale the water volume of inland waterbodies (Abstract PA0289). In *Proceedings of the 67th Congress of ISTAM* (pp. 1–3). Indian Society of Theoretical and Applied Mechanics.

Sharma, A., Vijay, S., & Ilampooranan, I. (2021). Glacial lake bathymetry derived from ICESat-2 and multispectral satellite images: A case study on Tso Lhamo Lake, Sikkim Himalaya. *Second International Conference on Natural Hazards and Risks in a Changing World 2021*.

F. Software and Open-source Datasets

Sharma, A., & Ilampooranan, I. (2026). Small and medium-sized inland waterbodies: Water volume predictions and flood implications [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.18218644>

Sharma, A., Narayanan, M., & Ilampooranan, I. (2026). An automated data-efficient morphometric approach to define global lentic and lotic inland waters [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.18277872>

Narayanan, M., Ilampooranan, I., & **Sharma, A.** (2025). Machine-Learning-Model-Builder [Computer software]. Zenodo. <https://doi.org/10.5281/zenodo.14907004>

Narayanan, M., Ilampooranan, I., & **Sharma, A.** (2026). Croplizer: An agro-socio-edapho-climatological tool for farmers [Computer software]. Hugging Face. <https://mukundn1997-croplizer.hf.space/>

9. Tools Developed (Open source)

ADEMA	Automated Data Efficient Morphometric Approach: Algorithm classifying global lentic and lotic inland waters down to 0.09 hectares (https://doi.org/10.5281/zenodo.18277872)
C-DAS	Cascaded Depth and Storage models: ML and DL models framework for estimating bathymetric features, including mean depth and volume of global inland waterbodies (<i>release planned following publication</i>)
StoRI	Storage and Reachability Index: A novel Index quantifying storage and reachability tradeoffs within the watersheds (<i>release planned following publication</i>)
TOI	Tankshed Overflow Index: A novel index developed to identify the location of new waterbodies in the watershed (https://doi.org/10.5281/zenodo.18218644)

10. Research Supervision and Leadership Experience

Research Mentoring: Provided research guidance to three Master's students (2 national, 1 international) working on machine learning applications in water resources and agricultural management. Activities included research design and methodological development, implementation of machine learning models and data analysis, and feedback during thesis and manuscript preparation.

11. Teaching Experience

Teaching Assistant Experience

- Water Resources Planning and Management (2024)
- Sustainable Water Resources (2023)
- Soil and Agronomy (2022)
- Crop System Modeling (2021)
- Environmental Laws and Legislation (2019)

12. Awards and Honors

2026-2027	EGU26 Travel Grant Travel support to attend the European Geosciences Union General Assembly 2026
2023-2024	IEEE GRSS Travel Grant Travel support for participation in the IEEE International Geoscience and Remote Sensing Symposium 2023
2023-2024	DORA Travel Grant Institutional funding support for participation in the IEEE International Geoscience and Remote Sensing Symposium 2023
2020-2025	MHRD Fellowship Doctoral fellowship providing financial support for Ph.D. research from the Government of India
2021-2022	MMPY Incentive One-time incentive for Ph.D. research from the Government of Himachal Pradesh, India

2020-2021	CO-PREPARE Travel Grant Funding support for participation in the NatRisk Conference 2021
2018-2020	GATE Fellowship Fellowship supporting MTech studies from the Government of India
2013-2017	SJM Merit Scholarship Scholarship awarded for the Bachelor's studies from Satluj Jal Vidyut Nigam Limited, India

13. Peer Review Activities

- Earth's Future (Publisher: American Geophysical Union / Wiley)
- Environmental Research Letters (Publisher: IOP Publishing)
- Journal of Water and Climate Change (Publisher: IWA Publishing)

14. Professional Memberships

- Member, American Geophysical Union (AGU)
- Member, European Geosciences Union (EGU)
- Member, IEEE Geoscience and Remote Sensing Society (GRSS)
- Member, International Association of Hydrological Sciences (IAHS)

15. Conference and Professional Service

Administrative and Institutional Roles

- Member, Website Development Committee, Department of Water Resources Development and Management, IIT Roorkee (2023-2025).

Organization of Scientific Conferences

- Organizing committee member, International Conference on Climate and Weather-Related Extremes (ICCWE-2022).
- Organizing committee member, Roorkee Water Conclave (RWC) for the years 2022, 2024, and 2026.